Satellite Systems

Introduction

In today's community exists a growing demand and necessity for really global voice and data communication.

To provide connections for these applications, two main technologies are in use today:

- Direct cable/fiber connections
- radio transmission.

Introduction

Cables are (at least for long distances, and especially between continents) expensive to deploy and hard to maintain, and they cannot reach every desired position.

Terrestrial radio links usually need to be in direct line of sight of each other, which makes them only usable over relatively short distances and requires many relay stations for longer distances.

A possible way to overcome these problems and to achieve real global coverage is the use of satellites.

Categories of Satellite Systems

Satellites can be divided into three different categories.

- Geostationary Earth Orbit Satellites
- Medium Earth Orbit Satellites
- Low Earth Orbit Satellites

Geostationary Earth Orbit

Geostationary orbits are circular orbits that are orientated in the plane of the earth's equator.

In a geostationary orbit, the satellite appears stationary, i.e. in a fixed position, to an observer on earth.

Technically, a geostationary orbit is a circular orbit in the equatorial plane with an orbital period equal to that of the earth.

This is achieved with an orbital radius of 6.6107 (equatorial) earth radii, or an orbital height of 35786 km.

Geostationary Earth Orbit

Geo satellite applications are best suited to carry large volumes of communications traffic.

Virtually all today's modern financial business is conducted at high speed via satellite example being credit card transactions, corporate retail inventory, shipment tracking, etc.

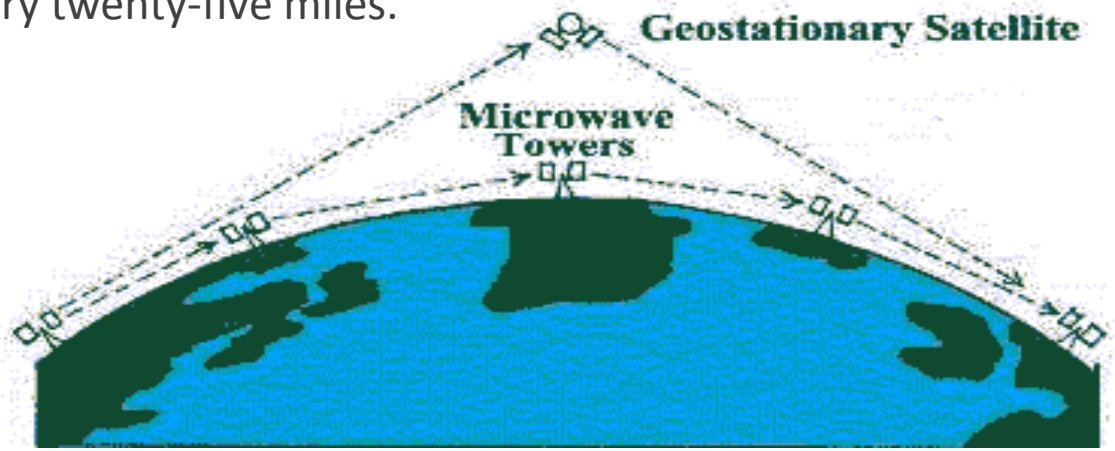
Geo satellites rapidly and efficiently link remote areas of the Earth with telephone, television and data information and provide transmission from a single uplink location to provide reception to a regional downlink pattern - this is often called point to multi-point distribution.

Satellites utilize transmit/receive signals in the microwave frequency range. They are not deflected by the Earth's atmosphere as lower frequencies are. Basically, they travel in a straight line, known as line of sight communication.

Geostationary Earth Orbit

Before the development of geo satellite technology, microwave transmission was accomplished via a series of repeater microwave towers and subsea cables.

The obvious limitation of microwave towers is they are a point-to-point technology and limited to line-of-site in regards to their placement - a tower is required every twenty-five miles.



Geostationary Earth Orbit

Example:

- The ***Inmarsat Satellite System*** is an example of using a geo orbit global system to provide global communication services to earth users.
- Its complete earth coverage (except for the poles) is achieved through four geo satellites.

Medium Earth Orbit

MEO s are circular orbits at an altitude of around 10,000 km.

Their orbit period measures about 6 hours.

The maximum time during which a satellite in MEO orbit is above the local horizon for an observer on the earth is in the order of a few hours.

Medium Earth Orbit

A global communications system using this type of orbit, requires a modest number of satellites in 2 to 3 orbital planes to achieve global coverage.

MEO satellites are operated in a similar way to LEO systems. However, compared to a LEO system, transfer of information from one satellite to another is less frequent, and propagation delay and free space loss are greater.

Medium Earth Orbit

The United States Navstar Global Positioning System (GPS) is a prime example of a MEO system.

It is a space-based triangulation system using satellites and computers to measure positions anywhere on earth.

GPS is a radio navigation system developed and operated by the U.S. Department of Defense.

It allows land, sea, and airborne users to determine their three dimensional position, velocity, and time, 24 hours a day, in all weather, anywhere in the world.

The uniqueness of this navigational system is that it avoids the limitations of other land-based systems such as limited geographic coverage, lack of continuous 24-hour coverage, and the limited accuracies of other related navigational instruments.

Low Earth Orbit

There are two classes of low earth orbit satellites:

- Little LEOs:
  - They are used for non-real time communications. Applications for these are data transfer like paging or electronic mail.
  - Little LEO satellites are quite small with a weight of 50-100 kg.
- Big LEOs:
  - Provide voice transfer (cellular telephony) or networking. They are built with a weight of up to 500 kg and a diameter of not more then a few meters.

Low Earth Orbit

Another aspect to classify satellites is the frequency of their transmissions:

- 800 MHz for little LEOs
- 2 GHz for big LEOs
- 20-30 GHz for broadband LEOs.

Low Earth Orbit

LEOs are usually located from 700 km from the earth to 1500 or 2000 km.

The low altitude of the orbit makes the transmission time for a link between a ground station and the satellite much shorter than it is for geostationary satellites.

Another advantage of low altitude is that less power is needed for the connection which in turn means that antennas used at the ground stations can be very small.

So it is possible to use a device not bigger than an cellular phone which is quite important if the network should be used for things like mobile telephony.

Low Earth Orbit

The velocity of low earth orbit satellites moving around the globe is about 25000 km/h, so the orbital period is around 100 min, depending on their height.

This also means that a LEO satellite is visible from a single point on the earth only for a short period of time (3-12 min).

To achieve global coverage a high number of satellites is required.

Most LEO satellite systems use polar orbits, each with a certain number of satellites.

Due to the earth's rotation beneath them, every satellite covers the whole surface within several orbital cycles.

Low Earth Orbit

The area which is covered by a single satellite with a certain minimum elevation angle taken in consideration is called its footprint.

The footprint between two adjacent satellites in each orbit is overlapping, and so are the footprints of two adjacent orbits.

Low Earth Orbit

As LEO networks are basically just another kind of digital data transfer network, all the general problems of routing in these structures are also present in satellite networks.

For a satellite system covering the whole globe it's important to have a reliable routing protocol adapted to the special needs of wireless satellite communication.

This includes the connection between earth stations and satellites as well as inter-satellite links (ISL).

A satellite needs an uplink and a downlink and at least two inter-satellite links. This is the minimum number of links needed for reasonable connectivity on the network.

Low Earth Orbit

When a satellite is leaving the visible area of its client during an existing connection, this connection must be given to another satellite.

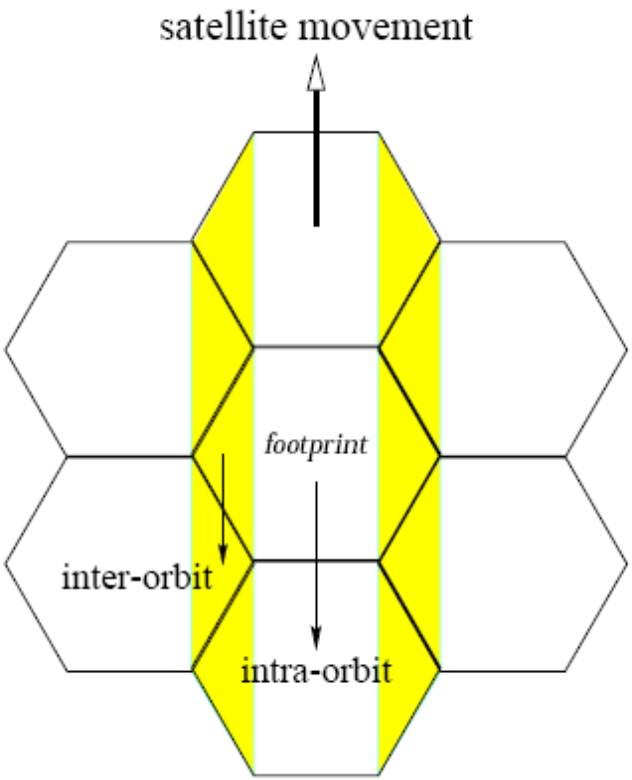
This process is called **handover** and requires **rerouting** of intersatellite links.

There are two different kinds of rerouting of intersatellite links:

- intra-orbit handover to the next satellite on the same orbit,
- interorbit handover to one of the satellites covering the next orbit.

- Both can happen while the satellite is moving around the globe.

Low Earth Orbit



footprint and handover

The Iridium System

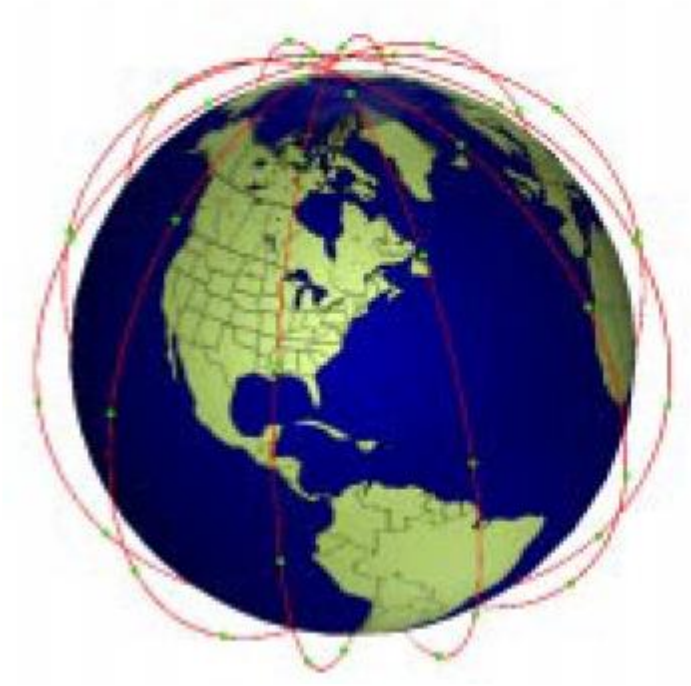
The Iridium system, which is the first operational LEO communication network, is a voice telephony and slow data connection network.

The space segment of the Iridium system consists of 66 operational plus 6 spare satellites in 6 circular near-polar orbits 785 kilometers above the earth's surface, giving the system the possibility of true global coverage.

Every Iridium satellite has 48 transmission beams, leading to a total number of possible 3,168 beams (of which only approximately 2,150 will be active at any given point in time, because some beams are switched off above the poles to avoid confusion due to beam overlap).

These beams create a grid of cells with a diameter of 600 km on the surface of the planet, analogical to a standard cellular phone cell network pattern.

The Iridium System



Iridium constellation structure

The Iridium System

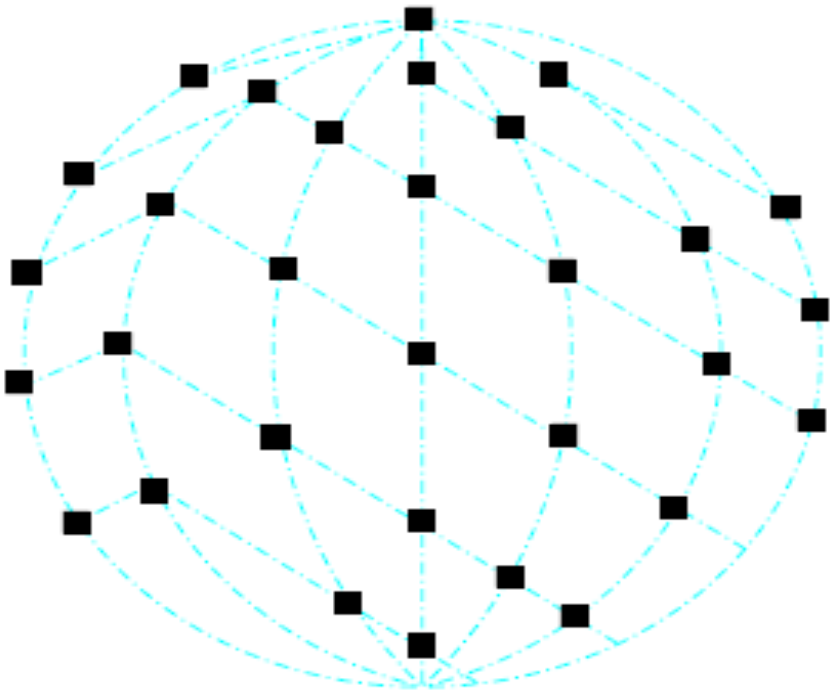
The cells move across the surface while the satellites circle above, making it necessary to hand over ongoing calls to the next satellite as the cell leaves the user.

Each Iridium satellite maintains inter satellite radio links to it's 4 nearest neighbours.

Routing of calls inside the network is done by the satellites themselves, making the space segment of the system independant in this aspect from ground control stations. Connection with the standard Public Switched Telephone Network (PSTN) is achieved through earth stations.

Theoretically, one such station would be enough for the connection of the satellite network with the standard PSTN, but for reasons of efficiency and cost reduction, about 15 to 20 stations in several different countries are planned.

The Iridium System



Iridium ISL pattern

The Iridium System

The frequencies used by the Iridium system are located in two different frequency bands.

The Ka-Band is used for the bandwith-craving connections in the system: the intersatellite links (23.18- 23.38 GHz) and the up- and downlinks to the PSTN gateways (29.1-29.3 GHz/19.4-19.6 GHz).

The L-band is used for the (much less bandwidthintensive) direct connection to the mobile end user devices (1,616-1,626.5 MHz).

To use the given relatively small ground-to-satellite bandwidth more efficiently, the system uses a combination of timedivision multiple access (TDMA) and frequency division multiple access (FDMA) methods for connections with the mobile units.

This allows for up to eight subscribers to use the same channel.

The Iridium System

The data rate for end users is 2.4kbps for voice, fax and data communication, and each satellite is able to handle between 1100 and 3840 simultaneous voice channels.

These Figures show clearly that the Iridium system is not primarily meant to be a data communication network (like e.g. the Teledesic system).

Additional services offered by Iridium will include paging and global positioning information. Iridium is available since the 1st of November 1998 everywhere on this planet, except for Cuba (on request of the US state department).